

No Logarithmic-Moment Criterion for Arbitrary C_0 -Semigroup Lévy Convolutions

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, for human review. The logarithmic moment condition is neither necessary nor sufficient for general strongly stable C_0 -semigroups. Necessity fails even for a nilpotent exponentially stable contraction, and sufficiency fails even for deterministic Lévy drift and a bounded strongly stable contraction generator. Exact drift and Gaussian tests, an iid-block reduction, and the sharp two-sided exponential regime are also proved.

1 Source question

The source is Zbigniew J. Jurek, *On Relations Between Urbanik and Mehler Semigroups*, arXiv:0811.2989 (2008) [1]. On printed page 12 it asks whether, for a C_0 -semigroup $(T_t)_{t \geq 0}$ on a separable Banach space and an E -valued càdlàg Lévy process Y , there is an analogue of the bounded-generator criterion

$$\int_{(0,t]} e^{sV} dY(s) \text{ converges as } t \rightarrow \infty \iff \mathbb{E} \log(1 + \|Y(1)\|) < \infty,$$

or, more generally, how convergence depends on the infinitesimal generator and on $Y(1)$. The exact source passage is reproduced below.

functions $t \rightarrow \widehat{\rho}_t(y)$ are differentiable at zero. The same setting is also in Fuhrman and Roeckner (2004).

Let us also recall here that stochastic processes, from Theorem 1,

$$U_t := T_t x + \int_{(0,t]} T_{t-s} dY(s), \quad t \geq 0,$$

are called *Ornstein-Uhlenbeck processes* and are well studied on Hilbert spaces. [These are solutions to so called Langevin equations.] Furthermore, one can easily express Lévy-Khintchine formula of U_t in terms of the corresponding parameters of $Y(1)$; cf. for instance Jurek-Mason (1993), Section 3.6 in Chapter 3.

Open problem. It is known that on arbitrary separable Banach space if $\lim_{t \rightarrow \infty} \exp tV = 0$ (in the operator topology; V is a bounded operator) then, for a càdlàg Lévy process Y , the limit $\lim_{t \rightarrow \infty} \int_{(0,t]} e^{sV} dY(s)$ exists (almost surely, in probability or in distribution) if and only if $\mathbf{E}[\log(1 + \|Y(1)\|)] < \infty$, cf. Jurek (1982).

Is there an analogous criterium true for C_0 -semigroups $\mathbf{T}$ on E ? Or how does the existence of a limit depend on the infinitesimal generator J of the semigroup $\mathbf{T}$ and the variable $Y(1)$?

Recall that Applebaum (2005) in Theorem 9, showed that log-integrability is sufficient for *exponentially stable contraction semigroup* $(T_t, t \geq 0)$ on a Hilbert space. His proof uses the Lévy-Itô decomposition of the process Y .

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The paper defines “operator topology” to mean the strong operator topology. That convention matters: one of the examples below also shows that the bounded-generator assertion immediately preceding the open question cannot be correct with merely strong convergence. A later review states the classical iff with convergence in *operator norm* [2, Section 6.1].

2 Answer and proof idea

There are two independent obstructions.

- A C_0 -semigroup may erase all old noise in finite time. Then no tail moment of the jumps is necessary.
- Strong stability may be arbitrarily slow in different directions. A driver with the best possible moment behaviour can then accumulate in infinitely many slowly decaying coordinates.

The first phenomenon is realized by the nilpotent translation semigroup on $L^2(0,1)$ and heavy-tailed compound-Poisson noise. The second is already realized without randomness: on ℓ^2 , let $T_t e_n = e^{-t/n} e_n$ and drive the convolution by the deterministic Lévy process $Y(t) = t(1/n)_n$.

The correct structural replacement is a block series. Writing the convolution over successive unit intervals produces iid variables ξ_n and the exact identity

$$\int_{(0,N]} T_s dY(s) = \sum_{n=0}^{N-1} T_n \xi_{n+1}.$$

Thus the generator contributes its discrete orbit (T_n) , while the law of $Y(1)$ contributes the one-block infinitely divisible law. This reduction gives closed criteria for deterministic drift and Gaussian noise. It also shows exactly when a logarithmic test returns: an exponential upper estimate gives sufficiency, and an exponential lower estimate gives necessity.

3 The logarithmic condition fails in both directions

Theorem 1 (Two-sided failure). *There are separable Hilbert spaces, strongly stable contraction C_0 -semigroups, and càdlàg Lévy processes for which each of the following holds.*

- (i) *The improper convolution converges almost surely although $\mathbb{E} \log(1 + \|Y(1)\|) = \infty$. The semigroup may be chosen nilpotent and hence exponentially stable.*
- (ii) *$Y(1)$ is deterministic (hence has every finite moment), but the convolution norm tends to infinity. The generator may be chosen bounded and the semigroup strongly stable and contractive.*

Consequently no logarithmic-moment iff criterion extends to arbitrary strongly stable C_0 -semigroups. Under the source paper's explicit strong operator convention, part (ii) also corrects its preceding bounded-generator formulation.

Proof. For (i), take $E = L^2(0, 1)$ and extend every $f \in E$ by zero outside $(0, 1)$. Set

$$(T_t f)(r) = f(r + t), \quad 0 < r < 1.$$

Translations on the zero extension show strong continuity; (T_t) is a contraction semigroup and $T_t = 0$ for $t \geq 1$.

Fix a unit vector $v \in E$. Let Y be a rate-one compound-Poisson process whose jumps are $R_k v$, where the positive iid amplitudes satisfy

$$\mathbb{P}\{R = e^m\} = \frac{6}{\pi^2 m^2}, \quad m \geq 1.$$

If N_1 is the number of jumps by time one, then on $\{N_1 = 1\}$ one has $Y(1) = Rv$. Therefore

$$\begin{aligned} \mathbb{E} \log(1 + \|Y(1)\|) &\geq \mathbb{P}(N_1 = 1) \mathbb{E} \log(1 + R) \\ &\geq e^{-1} \frac{6}{\pi^2} \sum_{m \geq 1} \frac{m}{m^2} = \infty. \end{aligned}$$

On the other hand, if S_k are the jump times, then

$$\int_{(0, t]} T_s dY(s) = \sum_{S_k \leq \min\{t, 1\}} T_{S_k}(R_k v).$$

This is a finite sum and is constant for $t \geq 1$, so it converges almost surely.

For (ii), take $E = \ell^2$ with standard basis (e_n) . Define

$$V e_n = -\frac{1}{n} e_n, \quad T_t = e^{tV}, \quad T_t e_n = e^{-t/n} e_n.$$

The operator V is bounded and (T_t) is a contraction semigroup. For each $x = (x_n) \in \ell^2$, dominated convergence gives

$$\|T_t x\|^2 = \sum_{n \geq 1} e^{-2t/n} |x_n|^2 \longrightarrow 0,$$

so $T_t \rightarrow 0$ strongly. It does not converge in operator norm, since $\|T_t\| = \sup_n e^{-t/n} = 1$.

Let $y = (1/n)_{n \geq 1} \in \ell^2$ and let $Y(t) = ty$. This is a deterministic Lévy process and $\mathbb{E} \log(1 + \|Y(1)\|) < \infty$. Its convolution is the Bochner integral

$$X_t = \int_0^t T_s y \, ds = (1 - e^{-t/n})_{n \geq 1}.$$

For every fixed t , this vector belongs to ℓ^2 because $1 - e^{-t/n} \leq t/n$. But monotone convergence yields

$$\|X_t\|^2 = \sum_{n \geq 1} (1 - e^{-t/n})^2 \longrightarrow \sum_{n \geq 1} 1 = \infty.$$

Thus the deterministic laws δ_{X_t} are not tight, and there is no convergence in distribution, in probability, or almost surely. $\square$

Remark 2 (Why exponential stability only repairs sufficiency). The nilpotent example satisfies $\|T_t\| = 0$ for $t \geq 1$, so even exponential stability cannot make logarithmic integrability necessary. The diagonal example has only strong stability; this is exactly why the usual exponential upper-bound proof of sufficiency is unavailable.

4 Exact generator criterion for deterministic drift

The deterministic witness is not an isolated trick; it is the complete drift case.

Theorem 3 (Range criterion). *Let (T_t) be a strongly stable C_0 -semigroup on a Banach space E , with generator A , and let $Y(t) = ty$. Then*

$$\int_0^t T_s \, dY(s) = \int_0^t T_s y \, ds$$

converges in norm as $t \rightarrow \infty$ if and only if

$$y \in \text{Ran}(-A).$$

If $y = -Ax$, the limit is x , and this $x \in D(A)$ is unique.

Proof. If $y = -Ax$ with $x \in D(A)$, the fundamental semigroup identity gives

$$\int_0^t T_s y \, ds = x - T_t x \longrightarrow x.$$

Conversely, suppose $x_t := \int_0^t T_s y \, ds \rightarrow x$ in norm. The integrated-semigroup identity says

$$x_t \in D(A), \quad Ax_t = T_t y - y.$$

Strong stability makes the right side converge to $-y$. Since A is closed, $x \in D(A)$ and $Ax = -y$. Finally, if $Az = 0$, then $T_t z = z$ for all t , while strong stability gives $T_t z \rightarrow 0$; hence $z = 0$. $\square$

For the diagonal generator in Theorem 1, $(-A)x = (x_n/n)_n$. The chosen $y = (1/n)_n$ would require $x = (1, 1, \dots) \notin \ell^2$, so $y \notin \text{Ran}(-A)$, exactly as the theorem predicts.

5 Exact iid-block reduction

Assume throughout this section that the finite-horizon stochastic integrals exist in E . Put

$$X_t := \int_{(0,t]} T_s dY(s).$$

Theorem 4 (Block-series criterion). *For $n \geq 0$, define*

$$\xi_{n+1} := \int_{(n,n+1]} T_{s-n} dY(s).$$

Then $(\xi_n)_{n \geq 1}$ is iid, with common law

$$\mathcal{L}(\xi) = \mathcal{L} \left(\int_{(0,1]} T_s dY(s) \right),$$

and, pathwise for every integer $N \geq 1$,

$$X_N = \sum_{n=0}^{N-1} T_n \xi_{n+1}. \quad (4.1)$$

If Ψ is the characteristic exponent of $Y(1)$, then the one-block law is determined by

$$\log \mathbb{E} e^{i \langle x^*, \xi \rangle} = \int_0^1 \Psi(T_s^* x^*) ds, \quad x^* \in E^*. \quad (4.2)$$

Suppose in addition that $T_n \rightarrow 0$ strongly and that the one-block convolution process has uniformly tight marginals on $[0, 1]$. Then convergence of X_t in probability as $t \rightarrow \infty$ is equivalent to convergence in probability of the series in (4.1).

Proof. The processes $u \mapsto Y(n+u) - Y(n)$ on disjoint unit intervals are independent and have the same law, which proves the iid assertion. Bounded deterministic operators commute with the stochastic integral, and the semigroup law gives

$$T_n \xi_{n+1} = \int_{(n,n+1]} T_n T_{s-n} dY(s) = \int_{(n,n+1]} T_s dY(s).$$

Summing proves (4.1). Formula (4.2) follows first for step-function integrands from independent increments and then for the defining finite-horizon integral by approximation.

For the last assertion write $t = N + r$, $0 \leq r < 1$. Then

$$X_t - X_N = T_N \xi_{N+1}(r), \quad \xi_{N+1}(r) := \int_{(N, N+r]} T_{s-N} dY(s).$$

The family $\{\xi_{N+1}(r_N) : N \geq 1\}$ is tight for every deterministic choice $r_N \in [0, 1]$. Strong convergence $T_N \rightarrow 0$ is uniform on compact subsets of E : this follows from the uniform boundedness principle and a finite-net argument. Hence $T_N \xi_{N+1}(r_N) \rightarrow 0$ in probability. Apply this to arbitrary real-time sequences and use the integer subsequence for the converse. $\square$

This is an exact general answer at the structural level. A C_0 -generator determines (T_t) uniquely, while $Y(1)$ determines the entire Lévy process law; equation (4.2) displays their interaction rather than reducing it to one radial moment.

6 The sharp exponential regime for a logarithmic test

Theorem 5 (Two-sided exponential criterion). *Let (ξ_n) be iid E -valued random variables. Suppose that for constants $c, C, \omega, \Omega > 0$,*

$$ce^{-\Omega n}\|x\| \leq \|T_n x\| \leq Ce^{-\omega n}\|x\|, \quad n \geq 0, \quad x \in E. \quad (5.1)$$

Then

$$\sum_{n \geq 0} T_n \xi_{n+1} \text{ converges almost surely} \iff \mathbb{E} \log(1 + \|\xi_1\|) < \infty. \quad (5.2)$$

Only the upper estimate in (5.1) is needed for the forward implication from the logarithmic moment to convergence; only the lower estimate is needed for the reverse implication.

Proof. Let $Z = \log(1 + \|\xi_1\|)$. If $\mathbb{E}Z < \infty$, then for every $\varepsilon > 0$,

$$\sum_{n \geq 1} \mathbb{P}\{Z > \varepsilon n\} < \infty$$

by comparison with the tail integral for $\mathbb{E}Z$. Borel–Cantelli gives $\|\xi_n\| \leq e^{\varepsilon n}$ eventually. Choose $0 < \varepsilon < \omega$; the upper bound in (5.1) then makes the series absolutely convergent almost surely.

Conversely, almost-sure convergence forces $\|T_n \xi_{n+1}\| \rightarrow 0$. The lower bound implies, eventually,

$$\|\xi_{n+1}\| \leq c^{-1} e^{\Omega n}.$$

If $\mathbb{E}Z = \infty$, comparison with the tail integral instead gives

$$\sum_{n \geq 1} \mathbb{P}\{\|\xi_{n+1}\| > c^{-1} e^{\Omega n}\} = \infty.$$

These events are independent, so the second Borel–Cantelli lemma says they occur infinitely often, a contradiction. $\square$

The two estimates isolate the counterexamples precisely. The nilpotent shift has no exponential lower bound; the diagonal semigroup has no uniform decay upper bound. If A is bounded, then $T_t = e^{tA}$ extends to a group and

$$\|T_n x\| \geq \|e^{-nA}\|^{-1} \|x\| \geq e^{-n\|A\|} \|x\|. \quad (5.3)$$

Thus a bounded generator automatically supplies the missing lower bound. If in addition $\|T_t\| \rightarrow 0$, the semigroup law upgrades this to an exponential upper bound.

For the block variable in Theorem 4, boundedness of A also makes T_s and T_s^{-1} uniformly bounded for $0 \leq s \leq 1$. Its Lévy measure is

$$\nu_\xi(B) = \int_0^1 \nu(T_s^{-1}B) ds.$$

The standard logarithmic-moment criterion for infinitely divisible laws therefore shows

$$\mathbb{E} \log(1 + \|\xi\|) < \infty \iff \mathbb{E} \log(1 + \|Y(1)\|) < \infty. \quad (5.4)$$

Equations (5.1)–(5.4) recover the classical bounded-generator criterion at integer times; the standard one-block maximal estimate gives its all-time form. In particular, they show why *operator-norm* stability and the bounded-generator lower estimate are the two decisive features missing from the general C_0 question.

7 Gaussian noise: an exact covariance test

For completeness, the other principal noise component has its own exact criterion. This is a standard Hilbert-space fact, included to make clear why a radial moment cannot encode the answer.

Proposition 6 (Gaussian energy criterion). *Let H be a separable Hilbert space and let $Y = W_Q$ be a Q -Wiener process, where Q is positive and trace class. Then*

$$\int_0^t T_s dW_Q(s)$$

converges in $L^2(\Omega; H)$ as $t \rightarrow \infty$ if and only if

$$\int_0^\infty \|T_s Q^{1/2}\|_{\text{HS}}^2 ds < \infty. \quad (6.1)$$

Proof. For $0 \leq u < t$, the Itô isometry gives

$$\mathbb{E} \left\| \int_u^t T_s dW_Q(s) \right\|^2 = \int_u^t \|T_s Q^{1/2}\|_{\text{HS}}^2 ds.$$

Thus the convolutions are Cauchy in $L^2(\Omega; H)$ exactly when (6.1) holds. $\square$

In the diagonal semigroup above, take $W_Q(t) = \sum_{n \geq 1} n^{-1} \beta_n(t) e_n$. Then $W_Q(1)$ is Gaussian and has every finite moment, but

$$\int_0^\infty \|T_s Q^{1/2}\|_{\text{HS}}^2 ds = \frac{1}{2} \sum_{n \geq 1} \frac{1}{n} = \infty.$$

More strongly, the convolution coordinates have variances

$$\sigma_n^2(t) = \frac{1 - e^{-2t/n}}{2n}.$$

Writing $M_t = \sum_n \sigma_n^2(t)$, one has $M_t \rightarrow \infty$ and

$$\text{Var}(\|X_t\|^2) = 2 \sum_n \sigma_n^4(t) \leq \frac{1}{2} \sum_n \frac{1}{n^2}.$$

Chebyshev's inequality yields $\|X_t\| \rightarrow \infty$ in probability. Hence even Gaussian noise with all moments can fail to have a limiting convolution under mere strong stability.

8 Literature boundary, limitations, and review recommendation

Applebaum's review [2, Theorem 6.3 and Section 6.1] records the known Hilbert-space Lévy-triplet characterization, attributed to Chojnowska-Michalik: Gaussian trace energy, integrated Lévy-measure mass, and a compensated-drift limit. It also records logarithmic integrability as sufficient under exponential stability, and the iff result for a contraction semigroup with bounded generator converging in operator norm. Those known results are consistent with every statement above.

The run's registry and attempts indexes were searched for arXiv:0811.2989 and for the central phrases. A bounded arXiv/web search through 17 August 2026 used the exact title, the exact question, " C_0 semigroup stochastic convolution logarithmic moment", "bounded generator", and

“Chojnowska–Michalik”. It found the source and Applebaum’s review, but no later paper explicitly claiming to answer Jurek’s general Banach-space question. Novelty is therefore plausible, not certified.

The packet does *not* claim a new arbitrary-Banach Lévy-triplet classification, nor does it claim the Hilbert triplet theorem as new. Its complete claim is the negative answer to a universal logarithmic analogue, the correction of the source’s strong-topology formulation, and the exact replacement statements proved here.

Human review should focus on three short points: the topology convention in the source, the block-law logarithmic equivalence (5.4), and the stated finite-horizon tightness condition in Theorem 4. The two counterexamples and Theorem 3 are elementary and independent of those auxiliary points.

References

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